The $p \neq q$ selfsimilar Besov multiplier question:
a sharp high-smoothness phase split and a counterexample

Autonomous solve-first investigation — lane 12

13 August 2026

Abstract

Schneider–Vybíral ask whether their identity $M(B^s_{p,p}) = B^s_{p,p,\mathrm{selfs}}$ for $0 < p \leq 1$ can extend to $p \neq q$. On the line, the answer is negative for every $0 < q < p \leq 1$ and $s \geq 1/p$. We construct one wavelet at each scale, placed near 2^j, with its coefficient normalized in $B^s_{p,q}$. Every rescaled unit localization sees at most one component, so the infinite sum belongs to $B^s_{p,q,\mathrm{selfs}}$; separated cutoff tests grow like $N^{1/p}$ before multiplication and $N^{1/q}$ afterwards, so it is not a multiplier. Combined with Nguyen–Sickel's multiplier theorem, this gives, for $s > 1/p$, equality when $p \leq q$ and strict containment when $q < p$.

1 Question and result

Fix the cutoff $\theta \in C^\infty_c(\mathbb{R})$ used in the source definition, and write

$$\|f\|_{B^s_{p,q,\mathrm{selfs}}} = \sup_{J\in\mathbb{N}_0,\, \ell\in\mathbb{Z}} \|\theta(\cdot - \ell) f(2^{-J}\cdot)\|_{B^s_{p,q}}.$$

Theorem 5.4(ii) of Schneider–Vybíral [1] proves $M(B^s_{p,p}) = B^s_{p,p,\mathrm{selfs}}$ for $0 < p \leq 1$; the following remark asks whether this can be generalized to $p \neq q$.

Theorem 1 (High-smoothness phase split)**.** *On $\mathbb{R}$ the following hold.*

1. *If $0 < p \leq q \leq \infty$ and $s > 1/p$, then*
$$M(B^s_{p,q}) = B^s_{p,q,\mathrm{selfs}} = B^s_{p,q,\mathrm{unif}}.$$
2. *If $0 < q < p \leq 1$ and $s \geq 1/p$, then*
$$M(B^s_{p,q}) \subsetneq B^s_{p,q,\mathrm{selfs}}.$$

In particular, the proposed $p \neq q$ generalization is false, including at the endpoint $s = 1/p$.

Part (1) is short. Nguyen–Sickel's Theorem 1.2 [2] says that $M(B^s_{p,q}) = B^s_{p,q,\mathrm{unif}}$ in this range. Taking $J = 0$ in the selfsimilar norm (and using the standard equivalence of smooth uniform localizations) gives $B^s_{p,q,\mathrm{selfs}} \hookrightarrow B^s_{p,q,\mathrm{unif}}$, while Theorem 5.4(i) of [1] gives $M(B^s_{p,q}) \hookrightarrow B^s_{p,q,\mathrm{selfs}}$. The three spaces therefore coincide. We prove part (2) explicitly.

2 A dilation-uniform separated wavelet sum

Choose a compactly supported orthonormal wavelet w with regularity and vanishing moments beyond s, and put

$$w_{j,k}(x) = 2^{j/2} w(2^j x - k), \qquad k_j = 4^j, \qquad c_j = 2^{-j}k_j = 2^j.$$

Let $\operatorname{supp} w \subset [-T, T]$, set

$$A = s + \frac{1}{2} - \frac{1}{p}, \qquad a_j = 2^{-jA}, \qquad m = \sum_{j\geq j_0} a_j w_{j,k_j}. \tag{1}$$

and take j_0 sufficiently large below. The supports

$$I_j = \operatorname{supp} w_{j,k_j} \subset [c_j - T2^{-j}, c_j + T2^{-j}]$$

escape to infinity and are disjoint, so (1) is a locally finite bounded function; indeed $a_j 2^{j/2} = 2^{-j(s-1/p)} \leq 1$.

Lemma 2. *If $s \geq 1/p$, then $m \in B^s_{p,q,\mathrm{selfs}}(\mathbb{R})$.*

Proof. After the dilation in the selfsimilar norm, the jth support is

$$2^J I_j \subset [2^{J+j} - T2^{J-j},\, 2^{J+j} + T2^{J-j}]. \tag{2}$$

The gap between consecutive intervals before dilation is $2^j - T(2^{-j} + 2^{-j-1})$. Choose j_0 so that this exceeds twice the diameter of $\operatorname{supp}\theta$. Since dilation only increases the gaps, every $\theta(\cdot - \ell)$ meets at most one interval in (2), uniformly in J and ℓ.

Suppose first that the surviving index satisfies $j \geq J$. Since

$$w_{j,k_j}(2^{-J}x) = 2^{J/2} w_{j-J,k_j}(x),$$

the wavelet characterization of $B^s_{p,q}$ and uniform boundedness of multiplication by translates of θ give

$$\begin{aligned} \|\theta(\cdot - \ell) a_j w_{j,k_j}(2^{-J}\cdot)\|_{B^s_{p,q}} &\lesssim a_j 2^{J/2} 2^{(j-J)(s+1/2-1/p)} && (3)\\ &= 2^{-J(s-1/p)} \leq 1. && (4) \end{aligned}$$

If $j < J$, the component has frequency below one. For every integer $0 \leq r \leq K$, where $K > s$, Leibniz' rule yields

$$\left\| D^r[\theta(\cdot - \ell) a_j 2^{j/2} w(2^{j-J} \cdot - k_j)] \right\|_\infty \lesssim a_j 2^{j/2} = 2^{-j(s-1/p)} \leq 1. \tag{5}$$

Here every derivative falling on w contributes $2^{j-J} \leq 1$, and derivatives falling on the fixed cutoff are harmless. The function in (5) has uniformly bounded support, so the standard $C^K_c \hookrightarrow B^s_{p,q}$ estimate gives a uniform Besov bound. Equations (4)–(5), together with the one-component property, prove the claim. □

3 The multiplier obstruction

Choose $\eta \in C_c^\infty(\mathbb{R})$ equal to one on $[-T, T]$. Enlarging j_0 if necessary, all translates $\eta(\cdot - c_j)$ and all of their supports enlarged by the translations occurring in an order-$r > s$ finite difference are mutually disjoint. For $N \geq 1$, define

$$g_N = \sum_{j=j_0}^{j_0+N-1} \eta(\cdot - c_j).$$

The difference characterization of Besov spaces then gives, with constants independent of N,

$$\|g_N\|_{B^s_{p,q}} \lesssim N^{1/p}. \tag{6}$$

Indeed the pth powers of the L_p norms add for the separated translates, both for the functions and for every finite difference.

The cutoff was chosen so that g_N selects exactly the indicated wavelet components of m. Hence

$$mg_N = \sum_{j=j_0}^{j_0+N-1} a_j w_{j,k_j}.$$

The wavelet characterization and the normalization of a_j imply

$$\|mg_N\|_{B^s_{p,q}} \asymp \left(\sum_{j=j_0}^{j_0+N-1} [2^{j(s+1/2-1/p)} a_j]^q \right)^{1/q} \asymp N^{1/q}. \tag{7}$$

If m were a pointwise multiplier, (6)–(7) would force $N^{1/q} \lesssim N^{1/p}$ for all N, impossible because $q < p$. Together with Lemma 2, this proves part (2) of Theorem 1.

4 Scope and verification notes

The construction is one-dimensional, which already disproves a general extension on $\mathbb{R}^n$. It works at $s = 1/p$: the right sides of (4)–(5) then remain constant rather than decay. The hypothesis $s \geq 1/p$ also places the source's difference-defined Besov space safely in the range where the standard Fourier/wavelet description agrees with it.

Nguyen–Sickel use the same separated translation indices in proving the necessity direction for localized multiplier classes. The additional step here is the uniform audit under every dilation in the 2012 selfsimilar norm, especially the broad-bump case $j < J$. Exact-phrase, arXiv-id, and selfsimilar-multiplier searches located no explicit answer to the source question. This is a bounded novelty check, not a priority claim.

Human-review recommendation. Accept as a candidate full negative answer in the stated parameter range. The key checks are the one-component property after arbitrary source dilations, the endpoint estimate (5), and the separated finite-difference bound (6).

References

[1] C. Schneider and J. Vybíral, *Non-smooth atomic decompositions, traces on Lipschitz domains, and pointwise multipliers in function spaces*, arXiv:1201.2280 (2012), Theorem 5.4 and the following remark.

[2] V. K. Nguyen and W. Sickel, *On a Problem of Jaak Peetre Concerning Pointwise Multipliers of Besov Spaces*, arXiv:1703.03246 (2017), Theorems 1.2, 1.6 and Proposition 3.10.